

House Price Prediction using Machine Learning

Prepared By

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Project Objective

The objective of this project is to predict house prices using machine learning techniques. The model estimates the price of a house based on various features such as area, number of bedrooms, bathrooms, parking spaces, furnishing status, and other amenities. This helps buyers, sellers, and real estate businesses make informed decisions based on data-driven predictions.

Dataset Information

The dataset used in this project is the Housing Prices Dataset obtained from Kaggle. The dataset contains information about different houses and their selling prices. The target variable is the house price, while the remaining columns are used as input features for prediction.

- Dataset Name: Housing Prices Dataset
- File Used: Housing.csv
- Number of Records: 545
- Target Variable: Price

Data Preprocessing

The dataset was cleaned before model training. Missing values were checked and handled appropriately. Duplicate records were removed. Categorical features such as main road access, guest room availability, basement, air conditioning, preferred area, and furnishing status were converted into numerical form using one-hot encoding.

Machine Learning Models Used

Two regression models were trained and compared:

1. Linear Regression
2. Random Forest Regressor

The dataset was divided into training data (80%) and testing data (20%) for model evaluation.

Visualizations

The following visualizations were created:

1. Histogram of House Prices
2. Correlation Heatmap of Features
3. Actual vs Predicted House Price Scatter Plot

These charts helped understand the distribution of house prices and the relationship between different features.

Key Findings

- Area is one of the most important factors affecting house prices.
- Houses with more bathrooms and better amenities generally have higher prices.
- Properties located in preferred areas tend to have greater market value.
- Furnishing status and air conditioning also contribute to price variation.
- Random Forest Regressor produced better predictions than Linear Regression.

Conclusion

This project successfully demonstrated the use of machine learning for house price prediction. After data preprocessing and model training, the Random Forest Regressor showed superior performance in predicting house prices. The results indicate that location, area, and amenities are major factors influencing property value. Such predictive models can assist real estate businesses in estimating fair property prices and making better investment decisions.